

Resistance, resilience, and community dynamics in mediterranean-climate streams

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Abstract Streams and rivers in mediterranean-climate regions (med-rivers) are subjected to sequential, yet contrasting hydrologic disturbances of drying and flooding. Although seasonally predictable, these disturbances can vary in intensity and duration within and among mediterranean-climate regions (med-regions). Consequently, med-rivers differ in the permanence of their aquatic habitats. To persist, species have acquired matched resistance and resilience adaptations. They gain resistance either by enduring the stress or avoiding it. Community recovery (or resilience) is achieved with cessation of hydrologic stress that permits maximization of re-colonization and reproduction. Endurance strategies are usually disturbance-specific, but avoidance enables organisms to cope with both drying and flooding, and is the prevalent resistance strategy. Correspondingly, community persistence depends to a large extent on the integrity of refuges, an aspect that has so far been little

explored. Existing information suggests that seasonal community succession becomes more pronounced with increasing aridity and declining water permanence. The invertebrate community in semi-arid med-rivers can therefore undergo succession through three to four identifiable assemblages, whereas in perennial streams the difference between wet and dry period assemblages is smaller. Community turnover is influenced by the intensity of the hydrologic disturbances and varies between wet and drought years.

Keywords Drying · Flooding · Invertebrates · Fish · Persistence adaptations · Community succession

Introduction

Resistance and resilience are responses by which organisms cope with stresses, and how populations and communities persist in disturbed ecosystems (e.g., Townsend & Hildrew, 1994; Fritz & Dodds, 2004). Fluvial ecosystems experience natural flow fluctuations (Lytle & Poff, 2004; Allan & Castillo, 2007) in the form of seasonal or a-seasonal floods and droughts (e.g., Poff, 1992; Lake, 2000, 2003). These hydrologic events are viewed as disturbances when they alter habitat structure and conditions, and modify resources, thereby threatening the persistence of populations through mortality-causing events (Resh et al., 1988;

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Townsend & Hildrew, 1994; Lytle, 2001; Stanley et al., 2010). Hydrological disturbances are characterized by their physical nature, such as their intensity, spatial extent, and frequency (Lake, 2000, 2003), timing and duration (e.g., Peterson & Stevenson, 1992), dynamics (e.g., pulse, press, or ramp; Lake, 2000), and their impact on stream communities (e.g., catastrophic or non-catastrophic; Platt & Connell, 2003). The impact of hydrological disturbances on the biota is also influenced by local geology and structural heterogeneity (Hart & Finelli, 1999; Lancaster, 2000; Magalhães et al., 2002; Pires et al., 2008; Boix et al., 2010), as well as by geomorphology and channel physiography (e.g., constrained vs. floodplain rivers; Lake, 2007). Hydrological disturbances can therefore act as environmental filters that impose strong selection forces on the biota (Poff, 1997). To persist in disturbed fluvial ecosystems, species have acquired matched adaptations that support the resistance and resilience of their populations (Lytle & Poff, 2004).

Global climate change (GCC) is predicted to increase the frequency and magnitude of extreme weather conditions (IPCC, 2007), thus threatening the survival of sensitive species (Hering et al., 2009; Jonsson & Jonsson, 2009; Domisch et al., 2011). Freshwater ecosystems are highly vulnerable to GCC, mainly because of the limited dispersal abilities of most aquatic biota, the direct impact of weather conditions on aquatic habitats, and the fact that many aquatic systems have already been severely impacted by human activities (Nickus et al., 2010; Woodward et al., 2010). Water-stressed areas, such as mediterranean regions (Mediterranean basin and parts of coastal California, Chile, South Africa, and South-West Australia; hereafter “med-regions”) are predicted to be particularly affected by GCC (Lake, 2000; IPCC, 2007; Ormerod, 2009; Lawrence et al., 2010). Streams and rivers in med-regions (hereafter “med-rivers”) are characterized by a relatively high share of endemic species (Bonada et al., 2007b; Hering et al., 2009), underscoring the threat of extinctions and biodiversity loss in these ecosystems (further discussed by Filipe et al., 2012).

In this article, we review the existing information on resistance and resilience adaptations, and the respective responses of invertebrates and fish, under contrasting hydrologic regime of drying and flooding in med-rivers. We also address the consequential changes in invertebrate community structure and describe their dynamics on an annual cycle.

Climatic settings and hydrology

Med-regions are climatically characterized by their relatively mild and wet winters and hot and dry summers (Aschmann, 1973). Annual rainfall varies within and among med-regions (e.g., Fig. 1), usually within a range of $>700 \text{ mm y}^{-1}$ in the relatively mesic (wet-mediterranean) regions, to $<500 \text{ mm y}^{-1}$ (but not less than 300) in the semi-arid (dry-mediterranean) ones (Cowling et al., 2005). In addition, annual rainfall within med-regions may vary markedly between years, and in an unpredictable succession (low constancy, Gasith & Resh, 1999), resulting in average, wet, or drought years (e.g., Bêche et al., 2006; Skoulidakis et al., 2011). This for example, can be demonstrated in the case of the Fuirosos catchment in north-eastern Spain, in which annual rainfall has varied by *ca.* 230% (480–1,070 mm), within less than a decade (Acuña et al., 2005).

The interplay between precipitation and temperature, and the respective loss of water (by evapotranspiration, Latron et al., 2009; and transmission losses) during the dry season, produces a predictable annual hydrologic pattern of seasonal, sequential drying and flooding that markedly vary in intensity among and within med-regions (Gasith & Resh, 1999; Bonada & Resh, this issue). This results in a variety of water permanence nuances, ranging from perennial to ephemeral situations (e.g., Fig. 2; Gallart et al., 2011). In mesic med-regions, or in karstic areas where base flow is regulated by large springs (e.g., Fig. 2A, Dan River, Israel, Degani et al., 1993; Bonada et al., 2007a), flow can consistently be perennial. In contrast, in drier med-regions, stream desiccation can be severe (e.g., Fig. 2B–D), resulting in cessation of flow and partial stream drying (i.e., hydrological discontinuity and formation of isolated pools, Gallart et al., 2011). The latter are often defined as intermittent, temporary, or non-perennial (Bêche et al., 2006; Bonada et al., 2007a; Sánchez-Montoya et al., 2007; Bond et al., 2008; Larned et al., 2010; Chester & Robson, 2011; Datry et al., 2011; Gallart et al., 2011; García-Roger et al., 2011; Steward et al., 2012). In extreme situations (i.e., during drought years, and as a result of human interventions) the entire channel network may dry-out during the dry period (Chester & Robson, 2011; Skoulidakis et al., 2011). Such streams have sometimes been categorized as temporary (e.g., Argyroudi et al., 2009; Larned et al., 2010; Chester & Robson, 2011; Datry et al., 2011; Steward et al., 2012) or

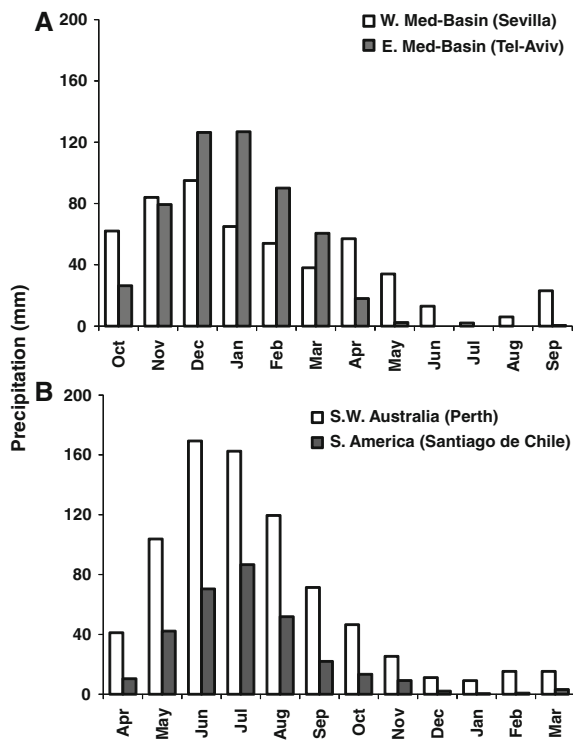


Fig. 1 Examples of precipitation variation in med-regions: **A** Localities representing the Eastern and Western Mediterranean Basin (Tel-Aviv, Israel and Seville, Spain, respectively) showing a similar mean annual rainfall (~ 530 mm), but different monthly distribution; **B** Localities representing med-regions in the southern hemisphere (Perth, South-West Australia, and Santiago de Chile, South-America) showing high divergence in mean annual rainfall (~ 790 mm and *ca.* 310 mm, respectively), but similar monthly rainfall distribution. Source: www.weatherbase.com; variation in precipitation records may exist between meteorological stations

ephemeral (e.g., Bonada et al., 2007a). However, the term ephemeral streams is usually used in relation to fluvial ecosystems with no base flow, such as washes or waddis in desert environments (Steward et al., 2012), in which discharge is episodic, occurring only during and immediately after precipitation events (corresponds to “episodic-ephemeral” in Gallart et al., 2011). We suggest distinguishing those streams from temporary med-rivers that partially dry-out in summer, but can retain base flow for several months, and classify the latter as “quasi-ephemeral” systems (corresponds to “intermittent-dry” in Gallart et al., 2011; e.g. Fig. 2C, D).

The degree and rate of channel drying can also vary spatially and temporally among wet and drought years (Fig. 2B–D), (Stanley et al., 1997; Boulton, 2003; Acuña et al., 2005; Bêche et al., 2006; Larned et al.,

2010). This can be demonstrated for example, in a third order med-river in north-eastern Spain that alternated between partial drying in dry years and perennially flowing in a wet year (<650 and $>1,000$ mm y^{-1} , respectively; Acuña et al., 2005).

Biotic adaptations and responses to hydrologic disturbances

Life-history adaptations

In recent years, a trait-based approach has become widely used for studying med-river ecosystems, thereby attempting to link community patterns with ecological processes (e.g., Usseglio-Polatera et al., 2000; Bêche & Resh, 2007; Bonada et al., 2007a, b; Díaz et al., 2008; Statzner & Bêche, 2010). The habitat template theory (Southwood, 1977, 1988) suggests that spatio-temporal variability of habitat heterogeneity is the setting against which organisms have evolved life-history strategies for maximizing fitness (Townsend & Hildrew, 1994). Life-history strategies are supported by suites of traits that determine the ability of an organism to cope with changing environmental conditions (Williams, 1985, 1996; Verberk et al., 2008). Lytle & Poff (2004) suggested that for each mode of adaptation, different components of the natural flow regime can be particularly important; for example, timing for life-history adaptations, predictability for behavioural adaptations, and magnitude and frequency of the disturbance for morphological adaptations. Moreover, extreme environmental conditions are likely to lead to the development of similar suites of traits (Statzner et al., 2001, 2004).

Resistance and resilience responses

Holling (1973) used the term “ecological resilience” in relation to the stability of ecosystem structure and function (reviewed in Peterson et al., 1998 & Gunderson, 2000). In the current scientific literature “resilience” is often used together with “resistance” for describing persistence of communities in disturbed ecosystems (e.g., Fritz & Dodds, 2004). When first introduced into stream ecology, resistance was defined as “the degree of community change caused by a disturbance” and resilience as “the rate of the

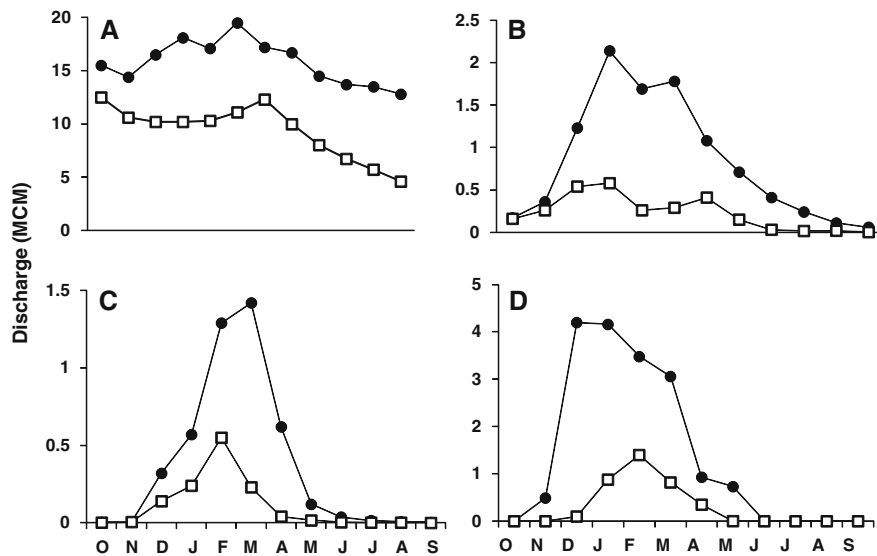


Fig. 2 Mean monthly discharge (MCM: million cubic meters) in med-rivers in Israel, on wet and dry years (hydrologic years 1992/1993 and 1998/1999; circles and squares, respectively). Four different hydrologic nuances of water permanence are shown: **A** perennial flow on both wet and dry years (Dan River);

B perennial flow on wet years and intermittent on dry years (Ga'aton stream); **C** intermittent flow on wet years and quasi-ephemeral on dry years (Ha'Shofet stream); **D** quasi-ephemeral flow on both wet and dry years (Saar stream). Source of hydrologic information: Israel Hydrologic Service

subsequent recovery” (Webster et al., 1983). Nevertheless, current use of these terms is often indistinct in that some authors use the term resistance for describing both the ability to withstand the disturbance as well as to avoid it (e.g., Townsend et al., 1997; Fritz & Dodds, 2004; Lake, 2007; Bond et al., 2008). The aerial life-stage of aquatic insects, for example, is sometimes considered a drought-resistant adaptation (e.g., Lytle, 2008) as well as a resilience strategy (e.g., Bêche et al., 2006; Robson et al., 2011). Likewise, desiccation-resistant life stages (e.g., resting eggs), which are clearly resistance adaptations to drying, are also viewed as traits supporting resilience by providing a pioneering advantage to the organisms that remain on site and can therefore rapidly re-colonize restored sites (e.g., Robson et al., 2011).

We adopted terminology that distinguishes between adaptations and responses that support resistance, from those mechanisms that promote resilience (Fig. 3). Accordingly, population resistance to hydrologic disturbance can be met by two main strategies: (i) endurance, i.e., tolerating the disturbance on site (i.e., within a reach), including cases in which organisms find refuge in microhabitats (e.g., under rocks or leaves, near the banks or the stream bottom, or in the hyporheic

zone); and (ii) avoidance, i.e., timely escaping the disturbed site to refuges in other locations. Avoidance can be met either by active (e.g., swimming) or passive (drifting) in-stream migration to alternative sites (internal refuges), or by emigration to suitable out-of-stream habitats (external refuges), whether actively (e.g., by flying) or passively (e.g., by phoresy). Avoidance can be regulated endogenously as part of the life cycle (Lytle & Poff, 2004) or evoked when stress conditions approach the organism's tolerance limit (Fig. 3).

Aquatic organisms can be grouped by the options of avoidance strategy they use for coping with hydrologic disturbances: (i) species that are entirely aquatic (e.g., worms, mollusks, most crustaceans, and fish), and are therefore forced to find refuge within the stream ecosystem (hereafter distinguished as “stream-confined” species); and (ii) species that can exit the water (e.g., by crawling on land or disperse by phoresy, and aquatic insects that have a terrestrial life stage) and are therefore able to exploit both internal as well as external refuge sites. Once stress conditions are relaxed, organisms that endured the disturbance can recover physiologically (Mas-Martí et al., 2010). Increases in population densities and community

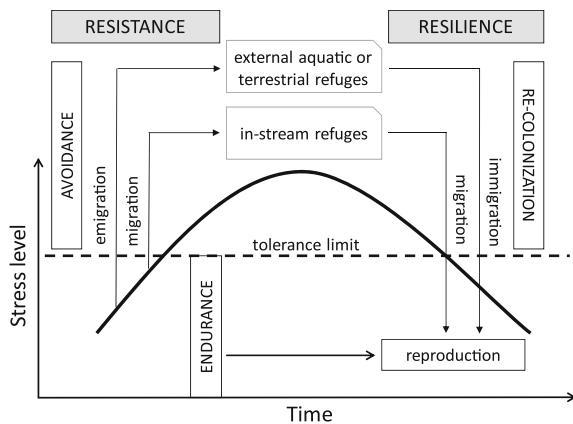


Fig. 3 A schematic model of the different biotic responses of resistance and resilience in med-rivers, in relation to a hydrologic stress level (*solid curved line*). An example is given for the endurance and the respective tolerance level (*horizontal dashed line*), and the threshold levels for avoidance and re-colonization (intersections with the *dashed line*). See text for further details

species richness are achieved by re-colonization of species that migrate from in-stream refuges or emigrate from external refuges, as well as by reproduction (Fig. 3).

In the following discussion, we describe the changes in habitat conditions caused by hydrologic disturbances in med-rivers, and address the adaptations and responses of med-river biota. We begin with the desiccation period of one hydrologic year, ending it by the dry period of the following hydrologic year (Fig. 4). We follow this hydrologic pattern to outline an expected annual succession of the invertebrate community in an intermittent med-river (Fig. 5).

Desiccation period

Desiccation stress

The intensity of stream drying is the main environmental factor that differentiates perennial med-rivers from intermittent and quasi-ephemeral streams. In the latter, drying may result in increased siltation (Boulton, 2003; Bonada et al., 2006; Dewson et al., 2007) and accumulation of organic matter (Acuña et al., 2005). Water quality often deteriorates, salinity (for example) increases, and water temperature and dissolved oxygen fluctuate daily. This pattern results in extreme and contrasting dissolved oxygen

conditions (i.e., super-saturation and hypoxia) (Aparicio & Sostoa, 1999; Pires et al., 2000; Boulton, 2003; Acuña et al., 2005; Williams, 2006; Dewson et al., 2007; Boulton & Lake, 2008). These stressful conditions usually develop gradually (Lake, 2003; Acuña et al., 2005), but their impact that limits the existence of the biota (Walters, 2011; Walters & Post, 2011) may be abrupt (i.e., a “stepped” response, sensu Boulton, 2003; Dewson et al., 2007; Magalhães et al., 2007).

Endurance to desiccation

Several physiological and morphological adaptations promote endurance of med-river species to drying. For example, certain invertebrates (Annelida, Chironomidae, Cladocera) have physiological adaptations that enable their survival under hypoxic conditions (e.g., oxygen-binding metalloproteins, Eriksen et al., 1996), while others (Coleoptera, Hemiptera and Diptera) endure the stress by respiring air, using special morphological adaptations (e.g., plastron or siphons). Indeed, invertebrates having low dissolved-oxygen requirements are prevalent in intermittent streams during drought (e.g., “group g” in Usseglio-Polatera et al., 2000; Pires et al., 2000; Acuña et al., 2005; Bonada et al., 2007a; Boulton & Lake, 2008). Many of these organisms are also able to acclimatize to other stressors that are associated with reduced flow (Pires et al., 2000; Lind et al., 2006; Lake, 2007), such as elevated temperature (Domisch et al., 2011) and high salinity (Díaz et al., 2008). In addition to the direct impact of reduced water quality conditions, aquatic organisms in isolated pools are also exposed to elevated predation pressure by congregated fish (Pires et al., 2000) and predatory invertebrates (e.g., Heteroptera, larvae of Odonata and Coleoptera) (Boulton, 2003; Acuña et al., 2005; Power et al., 2008; Walters 2011). In this connection, it was suggested that high crawling ability and armored body may support endurance by reducing the risk of predation (Power et al., 2008; Walters, 2011).

When desiccation persists, isolated pools may dry-out completely. Morphology (e.g., small size, soft body) and behavioral adaptations (e.g., crawling, borrowing) enable invertebrates to endure the drought by exploiting on-site refuges, such as moist sediment under rocks, detritus patches, and shallow depressions beneath woody debris (Chester & Robson, 2011; Robson et al., 2011; Walters, 2011), or by sheltering in the hyporheic zone (del Rosario

Fig. 4 An example of an intermittent med-river hydrograph (Ha'Shofet stream, Israel) on two consecutive hydrologic years (2002/2003 and 2003/2004, left and right, respectively). Four hydrological periods are shown: *drying* between mid-summer and early autumn (*a*), *rewetting* before flooding in early autumn (*b1*), subsequent winter flooding (*b2*), and a period of hydrological *stress relief* during spring and early-summer (*c*). Source of hydrologic information: Israel Hydrologic Service

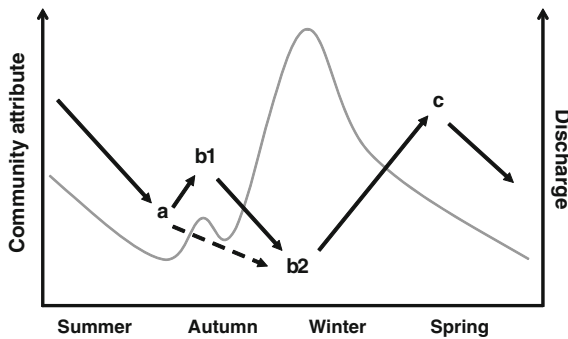
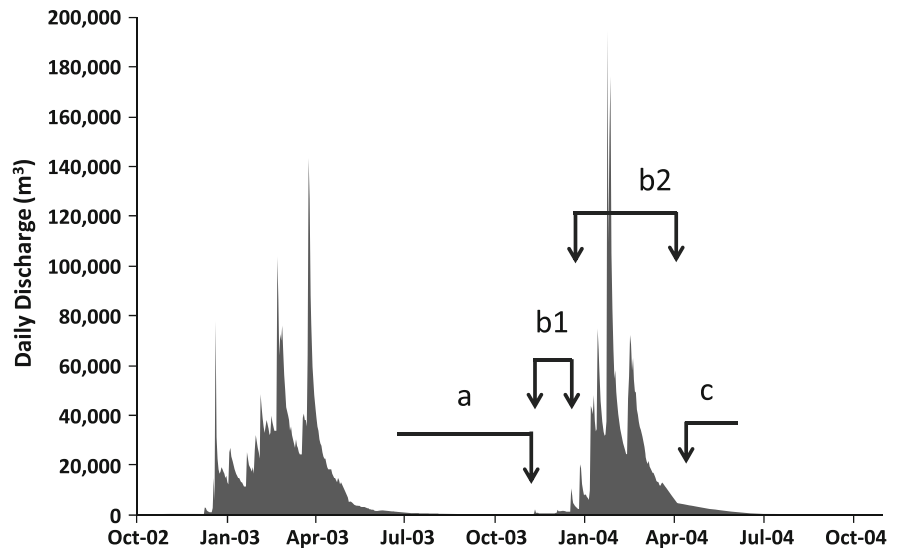


Fig. 5 Simplified schematic illustration of the direction (*black arrows*) of seasonal changes in invertebrate community attributes (e.g., richness, density) in an intermittent, semi-arid med-river, in relation to the annual stream discharge (*gray curve*). Letters indicate the successive assemblages likely to be formed in an average year: (*a*) a drought-resistant assemblage in late summer; (*b1*) a post-rewetting assemblage, formed in early autumn, before flooding; (*b2*) flood-resistant assemblage in mid-winter; and (*c*) a post-flooding assemblage, formed in spring-early summer, before desiccation. *Dashed arrow* represents a community change on years when first rains generate floods. The downward change in community attribute represents the degree of community resistance and the upward change represents resilience. For further discussion, see text

& Resh, 2000). Certain invertebrates (e.g., Ostracoda, Baetidae, Simuliidae, Ceratopogonidae, Chironomidae, and Plecoptera) can encyst or aestivate in the dry river bed (Delucchi & Peckarsky, 1989; Verberk et al., 2008; Chester & Robson, 2011; Robson et al., 2011; Walters, 2011). For instance, species of the stonefly family Perlodidae living in a temporary (i.e., quasi-ephemeral) med-river in southern Spain, are able to use different

strategies for surviving the dry season, including egg diapause, nymphal quiescence, and possibly using the hyporheic zone as a refuge (López-Rodríguez et al., 2009). Nevertheless, quiescent stages may still suffer mortality, as they may be subjected to predation by terrestrial and semi-aquatic arthropods (e.g., spiders, ants and riparian beetles; Hering & Plachter, 1997; Williams, 2006; Boulton & Lake, 2008).

Many med-river fish are habitat generalists that are able to survive both under lotic as well as lentic conditions (Mesquita et al., 2006; Morán-López et al., 2006; Mas-Martí et al., 2010). Fish living in intermittent streams (e.g., cyprinids; Ferreira et al., 2007), are also physiologically and behaviorally adapted to endure reduced water quality conditions that often develop at the time of desiccation (Magoulick & Kobza, 2003; Maceda-Veiga & De Sostoa, 2011). Nevertheless, during droughts, stranded fish often starve and their physiological condition deteriorates (Medeiros & Maltchik, 1999; Mas-Martí et al., 2010; Pires et al., 2010; Nagrodski et al., 2012), resulting in elevated fish mortality (Magalhães et al., 2007). In addition, congregated and weakened fish are more easily preyed upon by terrestrial predators, such as riparian birds (Bond et al., 2008) and mammals (e.g., otters; Magalhães et al., 2002).

Complete channel dehydration is intolerable by nearly all fish species (Morán-López et al., 2006; Skoulidakis et al., 2011). Only few fish with special physiological adaptations may endure a complete drying of the stream channel, but only for a limited

time. Certain species of catfish (Clariidae) for example, can respire aerially and survive on-site without surface water, as long as the sediment is moist (e.g., Sayer, 2005). However, as the dry season in med-rivers can be rather long (e.g., at least 5–6 months, as in semi-arid med-regions), the probability that fish will be able to survive in dry riverbed until the channel is re-wetted, is low. For this reason, quasi-ephemeral med-rivers are typically devoid of fish.

Avoidance of desiccation

Avoidance of disturbance is attained by the interplay between morphological adaptations that promote mobility (e.g., streamlined body, movement means that enable flying, crawling, swimming), adaptive behavior (e.g., rheotaxis, active and passive migration), and life-history traits such as short life cycle and synchronized emergence (Lytle & Poff, 2004). Fish for example, are highly mobile organisms that behaviorally respond to drying by in-stream migration to seek refuge in perennial reaches (Aparicio & Sostoa, 1999; Magalhães et al., 2003; Magoulick & Kobza, 2003; Davey & Kelly, 2007). A small number of fish (e.g., eels, catfish) and several invertebrates (e.g., Crustacea, Coleoptera, Heteroptera) can display amphibious behavior, migrating on the drying riverbed to wetted refuges (Sayer, 2005; Boulton et al., 2006). As previously discussed, stream-confined invertebrates can endure drying by encystment or aestivation, but some (e.g., Gastropoda; Pérez-Quintero, 2011) have been shown to survive desiccation by migrating up stream (by crawling or by phoresy). Amphipods and larvae of several insects (e.g., Diptera, Trichoptera, and Ephemeroptera) are able to respond to low-flow conditions by dislodging from the substrate and drifting with the flow (Delucchi, 1989; Dewson et al., 2007; James et al., 2009). In contrast, amphibious invertebrates are able to exploit both in-stream, as well as out-of-water, refuges (Lytle, 2008; Robson et al., 2011). Flight and aerial respiration combined with synchronized life cycles allow aquatic insect to optimize their inter-habitat strategy (Lytle, 2001; Lytle & Poff, 2004).

Formation of a desiccation-period invertebrate assemblage

Perennial med-rivers that maintain flow consistently may also retain relatively high overall habitat

heterogeneity during the dry period (e.g., the presence of riffles and pools; Bonada et al., 2007a). Consequently, the seasonal replacement of species in these streams can be relatively low. In contrast, drying in intermittent med-rivers can markedly alter habitat heterogeneity, imposing larger changes in community structure (Lind et al., 2006; Bonada et al., 2007a; Bêche et al., 2009; García-Roger et al., 2011). There is also a possibility for an intermediate situation between intermittent streams and perennial ones, for streams in which most lotic habitats were lost during drought but hydrological connectivity is still maintained (e.g., pools are connected to riffles; Bonada et al., 2006).

Reduced flow in med-rivers can lead to a local (reach scale) decline in richness and abundance particularly of rheophilic taxa (i.e., Ephemeroptera, Plecoptera, and Trichoptera—EPT; Boulton & Lake, 2008) by evoking avoidance response (Fig. 3) such as emergence of winged-adult insects or larval drift (Delucchi, 1989; Dewson et al., 2007; James et al., 2009). Alternatively, it can cause mortality from deterioration of habitat conditions (Acuña et al., 2005; Dewson et al., 2007). In contrast, newly formed or perennial lentic reaches can be colonized by species seeking drought refuges (i.e., Odonata, Coleoptera and Heteroptera—OCH; and certain Chironomidae; Acuña et al., 2005; Bonada et al., 2006, 2007a; Sánchez-Montoya et al., 2007; Lawrence et al., 2010). Overall density may increase locally, as these organisms congregate in isolated pools (e.g., Muñoz, 2003; Acuña et al., 2005; Lind et al., 2006; Dewson et al., 2007).

By the end of the dry period a desiccation-resistant assemblage is formed. In intermittent med-rivers it is usually dominated by lentic species (OCH and certain Chironomidae) that endure the stress, together with those that are filtered-in from the regional species-pool (e.g., Pires et al., 2000; Acuña et al., 2005; Robson et al., 2011).

Flood period

Flood stress

Drying in med-rivers may suddenly end with the onset of the rainy season in late autumn, when the channel is rewetted, and over time floods commence (e.g., Fig. 4, hydrologic year 2002–2003; Gasith & Resh, 1999;

Tarolli et al., 2012). Floods have multiple effects, they restore channel connectivity, homogenize water quality along the stream continuum (Tockner et al., 1999; Jenkins & Boulton, 2003), modify channel morphology (Grodek et al., 2012), redistribute accumulated sediments (Bourrin et al., 2008) and organic matter, scour encroaching vegetation (Corenblit et al., 2010), and redistribute the biota (Gasith & Resh, 1999).

Endurance to floods

Lentic invertebrates and fish that have survived the drying conditions in perennial streams and in permanent reaches of intermittent streams are subjected to the subsequent contrasting disturbance of strong scouring flows. Morphological adaptations (e.g., flat, streamlined or soft body, hooks, and claws; Bêche et al., 2006; Bonada et al., 2007) combined with behavioral responses such as moving near the banks or the stream bottom, hiding in crevices or under rocks, or occupying the hyporheic zone (low shear-stress microhabitats) enable certain invertebrates to endure high flows (e.g., *Gammarus pulex*; Franken et al., 2006). Larvae of Simuliidae for example, use hooks anchored to a secreted silk pad to attach to the substrate, and silk threads for moving between substrates. Simuliid species inhabiting fast flowing waters have more than tenfold higher numbers of hooks compared to species dwelling in slow velocities (Giller & Malmqvist, 1998). Simuliids have also been shown to respond behaviorally to increased flow by seeking shelter near the bottom, and returning to their original position once the flood has receded (Hart & Fonseca, 1997). Despite the adaptations that enable many stream invertebrates to maintain themselves during high flows, their community is not necessarily resistant to floods (Death, 2008).

Fish are able to withstand high flows by orienting their streamlined body toward the current and swim against it (positive rheotaxis; Mitchell, 1989). Alternatively, they can endure high flows by moving close to the stream bottom or near the stream banks, where the shear stress is reduced (Meffe, 1984; Pires et al., 2008).

Avoidance of floods

As in the case of desiccation, organisms persisting in med-rivers typically avoid the impact of flooding using mobility and morphological adaptations, combined with

behavioral and life-history traits (Townsend et al., 1997). Aquatic insects residing in streams that are impacted by seasonal floods can respond by synchronized pre-flood emergence (Lytle, 2001, 2002) or use environmental signals, such as rainfall or changes in flow velocity, to timely escape the strong currents (Lytle & White, 2007). For example, Gibbins et al. (2007) found that benthic invertebrates actively increase their entry into the drift in response to a threshold shear-stress under less than bank-full condition.

Formation of a flood-period invertebrate assemblage

We found less information on the effects of floods on med-rivers' invertebrate communities than on the effects of drying. Thus, for assessing the impact of floods on community structure and dynamics we also used information available for non-mediterranean streams. The impact of floods on stream communities is influenced by stream attributes such as channel morphology (constrained vs. floodplain streams; Lake, 2007), substrate heterogeneity (Lancaster, 2000; Brown, 2007; Pires et al., 2008) and stability (Imbert et al., 2005), and by environmental factors such as timing (e.g., spring or autumn; Merigoux & Dolédec, 2004) and magnitude (e.g. return time >50 years, Fritz & Dodds, 2004; >100 years, Vieira et al., 2004). Further discussion on the interactions of floods with these attributes is beyond the scope of present review.

Floods can reduce invertebrates' density by displacement and mortality (Hart & Fonseca, 1997; Gasith & Resh, 1999; Fritz & Dodds, 2004; Lake, 2007). A notable reduction in overall invertebrate density can also be caused by a "dilution effect" produced by the rising water level (Gasith & Resh, 1999). Invertebrates' density may also decline as a result of avoidance responses, such as a pre-flood mass-emergence of aquatic insects (Lytle, 2008) or larvae drift (Gibbins et al., 2007). Fritz & Dodds (2004) reported for example, that a major flood in a tall grass prairie stream (>50 year return time) reduced macroinvertebrates density by 99%. The effect of floods on community richness and composition is more ambiguous than their effects on density. Floods may locally eliminate lentic species (Merigoux & Dolédec, 2004), but re-colonization by rheophilic taxa (e.g., Trichoptera; Muñoz, 2003) may compensate for the loss of flood-intolerant ones.

Flooding in late autumn when the stream is at its lowest stage, is expected to have a stronger impact on communities residing in intermittent or quasi-ephemeral streams than on communities in streams that maintain perennial flow. In the latter, overall hydrological heterogeneity (e.g., runs, riffles, and pools) is maintained along the year, allowing coexistence of lentic and lotic species and thereby supporting relative stability of the community (Bêche et al., 2006; Bonada et al., 2006, 2007a). In contrast, early floods can markedly impact communities residing in intermittent streams, because late autumn assemblages consist mostly of drought-resistant lentic species (e.g., OCH and Chironomidae), which are more likely to be eliminated by strong flows. Overall, replacement of species by floods can lead to the formation of a flood-resistant assemblage (Argerich et al., 2004).

Evidence discussed above suggests that during the prolonged sequential, yet contrasting stress periods of drought and floods the invertebrate community in intermittent med-rivers can undergo succession of species comprising drought-resistant and flood-resistant assemblages. There is however a possibility for the formation of an additional short-term, but distinct assemblage, during the beginning of the wet season. This assemblage can be formed when first rains wet the channel (e.g., Fig. 4b1) and the hydrologic connectivity is partly or fully restored, but the onset of spates is delayed, until the soil in the drainage basin is saturated. At this time, the conditions in the rewetted stream are conducive for re-colonization, because of the seasonal drop in air and water temperature, the decreased salinity (from dilution by rainwater), and the high availability of organic matter and periphyton biomass (Acuña et al., 2007; Álvarez & Pardo, 2007). This pre-flood transitional period may last for several weeks (e.g., Fig. 4b1), during which invertebrates abundance and possibly richness may increase (Fig. 5b1).

The post-rewetting pre-flooding assemblage may consist of lentic species that have endured the drying period, including taxa that emerged from dormant stages (Robson et al., 2011), in-stream migrants, and insects that immigrated from external refuges. Acuña et al. (2005) reported the presence of a macroinvertebrate community following rewetting (post-drought) that was dominated by pioneering lentic Diptera. Supporting evidence for the existence of post-rewetting community also comes from findings of a study we conducted in an intermittent Med-river (Ha'Shofet

stream, Israel; Figs. 2C and 4). There, macroinvertebrates were collected by a kick-net (400 µm mesh size) on seven occasions (June 2003–February 2004) by disturbing the stream's bottom (mostly cobbles) along a 10 m stretch of a permanent reach. Bray–Curtis similarity analysis of abundance (family level) was performed using transformed data (square-root) on PRIMER 5 (PRIMER E Ltd., Plymouth, UK). From this analysis, we identified four distinct macroinvertebrate community assemblages (Fig. 6), two that were formed during the dry period (Fig. 4a; June–August and October/November) and two during the wet period (Fig. 4b; December/February). The early winter assemblage (December) was formed following rewetting of the stream, whereas the later (February) was formed after the commencement of floods, when monthly discharge increased by *ca.* 70-fold (Online resource 1, Fig. S1). Overall density of the dry period assemblage (November) differed from that of the post-rewetting assemblage (December), mostly contributed by the addition of rheophilic taxa. Gammaridae and Caenidae for example, increased by *ca.* 4-fold, Hydropsychidae and Simuliidae by *ca.* 5-fold, and Baetidae by *ca.* 25-fold; whereas density of Chironomidae larvae decreased by *ca.* 12-fold (Online resource 1, Table S1).

Re-colonization of the stream during this short, pre-flood period is risky for both invertebrates and fish because of the high temporal variability in rainfall distribution in med-regions. Early floods may wash away organisms or, alternatively, they may experience a recurrent desiccation stress following rainfall arrest (i.e., a “false start”; Boulton, 2003). First rains are sometimes sufficient to generate floods (e.g., Fig. 4, hydrologic year 2002–2003), thus preventing the formation of a post-rewetting assemblage, and the formation of a flood-resistance community can be abrupt (Fig. 5a–b2).

Post-floods period

Recovery mechanisms (resilience)

Cessation of rains marks the beginning of the dry period in med-regions and the start of a major transitional period of hydrologic stress relief (Fig. 4c) that lasts until desiccation stress begins. In semi-arid med-regions for instance, this period may last up to four months during

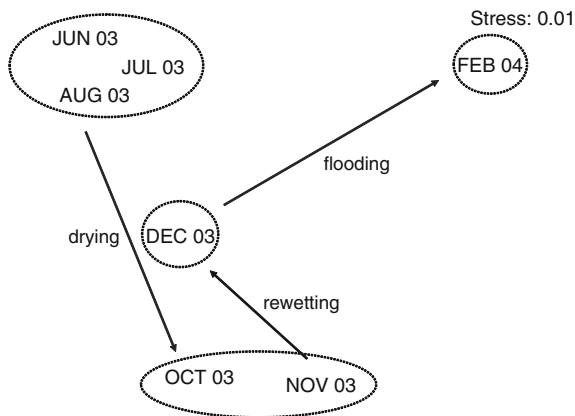


Fig. 6 Non-metric multi-dimensional scaling (NMDS) plot of macroinvertebrate assemblages (based on taxon density) collected in an intermittent stream (Ha'Shofet, Israel). Four distinct assemblages are identified: summer (June, July, and August 2003), autumn (October and November 2003), early winter (December 2003), and mid-winter (February 2004). For further details, see text

wet years. In both perennial and intermittent med-rivers, the environmental conditions during this period are benign, with moderate temperatures and flow velocities, high dissolved oxygen, and plentiful food resources (Gasith & Resh, 1999), and spatial habitat heterogeneity is highest. Species that sheltered in refuges during winter can now disperse and re-colonize the stream by actively migrating (e.g., fish; Pires et al., 2008), drifting with the current (e.g., invertebrates; Fritz & Dodds, 2004), or immigrating from external refuges (e.g., aquatic insects; Verberk et al., 2008). Phoresy of invertebrates by birds and mammals was suggested as another possible mode by which organisms can immigrate into streams (Green & Figuerola, 2005; Vanschoenwinkel et al., 2008; Waterkeyn et al., 2010), but the relative importance of this mode of dispersal is unclear (Bohonak & Jenkins, 2003).

Life-history adaptations in temporary streams promote reproductive plasticity that can reduce the constraints of the developmental period (Verberk et al., 2008; Larned et al., 2010). These adaptations include asexual reproduction, extended reproductive period, high fecundity, small body, short life cycle, and early maturity (Poff, 1997; Townsend et al., 1997; Statzner et al., 2004; Poff et al., 2006; Bêche & Resh, 2007; Bonada et al., 2007a, b; Dolédec & Statzner, 2008) which collectively promote multivoltinism (Bonada et al., 2007a; Díaz et al., 2008; Lytle, 2008). Likewise, synchronized reproduction can

reduce the risk of mortality by floods to early life stages (Pires et al., 2008). This is demonstrated for example, in the case of the Yarqon bleak (*Acanthobrama telavivensis*), a small endemic cyprinid that resides in the coastal region of Israel (Goren & Ortal, 1999). The fish initiates spawning toward the end of the flood period (February–April). This not only puts the early stages somewhat at risk of being dislodged by late floods (probability of late floods <25%; e.g., Fig. 4, hydrologic year 2002–2003) but also gains them a longer period of growth under favorable conditions (Elron et al., 2006).

Formation of post-floods, pre-desiccation invertebrate assemblage

The benign environmental conditions that prevail during the stress-free period, combined with restoration of the food base (Death & Zimmermann, 2005; Álvarez and Pardo, 2007) provide a window of opportunity for biological recovery of population densities and community richness and composition (Lake, 2003; Fig. 5b2–c). In the study of Ha'Shofet stream (Israel) described earlier, overall macroinvertebrate density in March and April increased by 5 to 10-fold, relative to their densities during the preceding flood period, which was mostly contributed by an increase of *Gammarus syriacus* (>7-fold) and *Baetis* sp. (>60-fold) (Online resource 1, Table S1).

Food-web relationships suggest that re-colonization and establishment of invertebrate populations will occur in a predictive succession following a progressive build-up of resource availability (Mackay, 1992; Álvarez & Pardo, 2007; Sabater et al., 2008; Álvarez & Pardo, 2009; Power et al., 2008). Taxa that endured flooding (i.e., have survived on-site), as well as pioneer species have the advantage of being the first to occupy restored habitats and exploit the rich food resources (Power & Dietrich, 2002). Filter-feeders that exploit particulate suspended matter (e.g., Simuliidae, net-spinning Trichoptera) are expected to be early colonizers, alongside or followed by, grazers and scrapers (e.g., some Ephemeroptera and Trichoptera) that consume periphyton, and by shredders (e.g., some Diptera and Plecoptera) that feed on conditioned coarse organic matter. Predators (e.g., Odonata and some Coleoptera), perform better when the stream level and flow velocity are reduced and prey density is high. Therefore they are likely to be relatively more

prevalent toward the end of the recovery period, and more so during the desiccation period (Acuña et al., 2005; Bogan & Lytle, 2007). In the extreme case of floodless years (e.g., Power et al., 2008), rewetting of intermittent streams may allow a limited, short-term recovery of populations (mostly of lentic species) that may abruptly end by rapid, subsequent drying.

Little is known on invertebrate community structure and dynamics under the extreme intermittency of quasi-ephemeral med-rivers. In the absence of fish, predation pressure on invertebrates is reduced. This suggests a rapid build-up of community species richness and population densities post-flooding. Similar to other temporary water bodies, quasi-ephemeral med-rivers are characterized by invertebrates having desiccation-resistant stages (Williams, 1985) and by aquatic insects that emigrate from external refuges. For example, Argyroudi et al. (2009) reported dominance of insects possessing desiccation-resistant eggs (e.g., Ephemeroptera, Plecoptera) in quasi-ephemeral med-rivers in north-eastern Greece. Community turnover in these ecosystems is expected to be mostly limited to a shift from a winter assemblage to that of a short-lived spring assemblage, until surface water disappears.

Conclusions

Resistance and resilience of communities in med-rivers

In this closing part, we highlight present insights on key characteristics of resistance, resilience, and community dynamics in med-rivers. Med-rivers around the globe are governed by a distinct pattern of seasonal rainfall distribution (usually rainless summers followed by wet winters), which greatly vary in amount, duration, and intensity among med-regions (e.g., semi-arid vs. mesic), as well as inter-annually within each region (e.g. dry vs. wet years). The seasonal pattern of drying followed by flooding is subjecting the biota to a relatively long period of stressful and contrasting hydrologic conditions, which in some med-regions, can extend for more than half of the hydrologic year. For surviving hydrological disturbances on site, some species use morphological adaptations, such as hooks or claws to endure strong flows, while others possess respiration traits, such as metaloproteins or plastron to

survive under low or no flow conditions. These endurance adaptations are usually stress-specific as for example aerial respiration that is mostly required for surviving drying conditions but less relevant for enduring floods (Bêche et al., 2006). Furthermore, some drought-resistant adaptations (e.g., armored body) may be detrimental at times of high flows (Power et al., 2008; Walters, 2011). Therefore, it is not surprising that avoidance, which promote resistance to both drying and flooding, is a primary resistance strategy of most med-river biota (e.g., Mesquita et al., 2006; Bonada et al., 2007a; Robson et al., 2011). This for example, is reflected by domination of aquatic insects in the invertebrate communities, particularly in intermittent and quasi-ephemeral med-rivers (Pires et al., 2000; Muñoz, 2003; Graça et al., 2004; Acuña et al., 2005; Bonada et al., 2007a; Sangiorgio et al., 2007; Chester & Robson, 2011). High mobility of winged-adults gains aquatic insects advantage over other groups of invertebrates, enabling them to use life-history or behavioral adaptations for escaping stressful conditions, as well as to rapidly re-colonize restored sites during the time of stress relief. Some aquatic insects (e.g., Perlodidae and Simuliidae) are also known to have desiccation-resistance stages (López-Rodríguez et al., 2009; Robson et al., 2011), which is advantageous for pioneering species. High mobility combined with reproduction (e.g., multivoltinism) support high recruitment of invertebrates and rapid establishment of their populations (high resilience).

The overall impact of drying and flooding on community structure and dynamics is expected to increase with declining water permanence (from perennial, to intermittent, and quasi-ephemeral), because of the respective greater change in habitat structure and availability along the year. Correspondingly, the seasonal formation of distinguishable drought-resistant, flood-resistant, and stress-free assemblages is expected to increase with declining water permanence. However, we hypothesize that along the declining wetness gradient of med-rivers, the change in the number of distinct assemblages formed on a yearly cycle is not linear, but rather hump-shaped. For different reasons, reduced turnover (fewer assemblages formed) is possible both in the wet as well as in the arid med-regions. In the former, the relatively stable conditions that prevail in perennial streams support higher proportion of species and traits common to both the wet and the dry periods (Bêche et al., 2006; Bonada et al., 2007a), whereas in the more

arid range (i.e., quasi-ephemeral streams), rapid and complete drying imposes existence of mostly a single community during a relatively short wet period. A high annual turnover of invertebrates in intermittent med-rivers is compatible with domination of the community by aquatic insects that are able to maximize avoidance strategies for coping with contrasting hydrological disturbances. In addition, the rate of community turnover is influenced by the intensity and duration of the hydrologic disturbances. In dry years for example, the transition from winter assemblage to that of spring-early summer is expected to begin earlier and develop more rapidly than on wet years (Gasith & Resh, 1999; Bêche et al., 2006).

Knowledge gaps and outlook

This review allows some insight into knowledge gaps and possible misconceptions regarding structure and dynamics of med-river ecosystem. For example, treating all med-rivers as a single, uniform group having a common hydrologic pattern of drying and flooding is only partly justified and to some extent problematic, because it obscures the diverse biological responses to hydrologic disturbances under different nuances of water permanency. Likewise, considering the impact of either drying or flooding on med-river communities may lead to overlooking of the sequential contrasting hydrologic stress, a central attribute of med-river ecosystems. We found limited information on the seasonal dynamic of invertebrate and fish communities in med-rivers. Also, relatively more information is available on the impacts of drying on community structure and dynamics in med-rivers than on the effects of flooding; the latter may partly result from a technical difficulty of obtaining information during high flows (Pires et al., 2000). Most of the information comes from limited “snapshot” sampling, once or twice a year, and therefore only dry and/or wet period assemblages are respectively identified (e.g., Muñoz, 2003; Bêche et al., 2006; Bonada et al., 2006; Lind et al., 2006; Boix et al., 2010; Chester & Robson, 2011; García-Roger et al., 2011). More frequent sampling is needed for describing the annual community turnover in different med-regions. As suggested by Robson et al. (2005) and reiterated by Lind et al. (2006), improvements in our understanding of variation in community structure requires more studies of patterns of natural variation in relation to different

water permanency situations, habitat diversity, and time scales. We also stress the strong need for conducting more studies on species-specific environmental requirements and thresholds, such as flow requirements (Buffagni et al., 2010), temperature and salinity preferences (Haidekker & Hering, 2007; Stamp et al., 2010; Schäfer et al., 2011), and dispersal limitation of aquatic biota (Bilton et al., 2001; Hering et al., 2009; Robson et al., 2011). These examples are expected to become more relevant under future climate change, particularly in med-regions (e.g., Grantham et al., 2010; further discussed by Filipe et al., 2012). Finally, evidence shows that the survival and recovery of many med-river species depends on the availability of permanent refuges (Bond et al., 2008; Larned et al., 2010; Pires et al., 2010; Chester & Robson, 2011; Robson et al., 2011). We therefore join others (e.g., Lawrence et al., 2010; Wilby et al., 2010; Robson et al., 2011) in underscoring the need to protect off-site refuges that are essential for maintaining the ecological integrity of med-rivers.

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